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Casewriter's Comments on

# FANSTEEL, INC.: THE ADVANCED STRUCTURES DIVISION

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## I. OBJECTIVES

The purpose of this note is to clarify the nature of the problems and opportunities facing Mr. Fletcher, the general manager of ASD, and to employ this analysis in arriving at a set of recommended courses of action.

The Fansteel case was written to serve as a teaching vehicle in the "implementation" portion of the Policy course. Normally, in this portion of the course, the strategy of the company is accepted as a given and discussion is limited to devising organizational forms and administrative actions that will serve to implement the strategy. In the Fansteel case, however, much of the complexity and confusion evident among management on organizational issues stems, it may be argued, from the vagueness and abstractness of the assigned strategy. Hence, an understanding of the limitations and interpretations of the strategy is a necessary prelude to intelligent implementation. For this reason, a portion of this note is devoted to a study of the strategy and the possible sub-strategies implicit within it. Throughout, the assumption is made that the strategy represents a goal that can possibly, although not necessarily readily, be achieved.

## II. SUMMARY OF THE PROBLEM AND THE REASONING USED IN THIS ANALYSIS

The Fansteel case describes the situation faced by the general manager of ASD in November 1967. For two years, 1965-66, Fletcher had been in charge of a division that consisted of only two important business activities: Fansteel/Torrance and Airtek. The manufacturing technologies used by these plants had enough in common that a central engineering group had been formed (p. 18). A division level marketing group had helped increase the plant's penetration of specific customers, and the scale of operations was small enough to allow Fletcher to personally act as the technical, marketing, and managerial coordinator among the different groups in the division. By November, 1967, however, ASD consisted of eight plants, with more acquisitions in sight. A centralized engineering group that could serve all the plants did not exist and few thought it was attainable. The division level marketing group did not have the capability of doing technical selling for the new plants and the scope of operations was too large for Fletcher to remain in contact with operations.

If full decentralization of business operations were acceptable, i.e., operation of a portfolio of fabrication businesses, there would be no major problem other than maintaining profitable operations. ASD's strategy, however, called for the "performance of a materials system management function," involving a number of boundary spanning activities. They were:

1. Bringing the scientific community and the fabrication businesses in closer contact.

2. Bringing the combined skills of the scientific community and the fabricators to bear on the materials problems of industry.

3. Developing methods for closely controlling fabrication activities that required inputs from several plants -- manufacturing coordination.

4. Developing new fabrication techniques by relating knowledge that existed in different plants -- coordinating engineers.

5. Representing the total combined capability to the customer.

It is not exaggeration to say that, somewhere in the case, someone points out the need for relating almost every activity within the division with every other activity, both within and without the division. [Perhaps it would be appropriate for ASD management to give thanks for the absence of international problems, lest the business become completely obscured by a thickening web of interrelationships!]

An analysis of the case could take as its goal the determination of the organization best suited to this complex strategy. It is easier, and probably more enlightening, to ask "What should Fletcher do?" in the context of the very real administrative constraints he faces and in terms of his own personal success as a division manager. With this approach in mind, the analysis was carried as far as seemed reasonable. A preview of line of reasoning used is given below:

1. ASD's "strategy" is actually more of a philosophy than a clear cut design for running a business.

2. The vagueness of the strategy permits a wide lattitude of interpretation.

3. The organizational designs proposed by members of management differ, not because they are seen as alternate ways to the same end, but because they are implicitly suggesting different sub-strategies for ASD.

4. The two basic sub-strategies that are suggested differ in their emphasis on technical versus managerial coordination. Some managers would prefer to concentrate on the prospects for ASD becoming a low cost supplier of complex structures by efficiently coordinating the manufacturing operations among plants. Others would prefer to emphasize the coordination of engineering and marketing skills in order to penetrate customers on the product design level.

5. These two sub-strategies imply different organizational structures and operating policies for ASD. The choice of one or the other would cause different managers to gain influence and commit the division to both the structure and the strategy.

6. Fletcher is uncertain as to which sub-strategy is more appropriate.

7. He is also, therefore, uncertain as to which structure to adopt.

8. Given that Fletcher's uncertainty is well founded, the proper question is "What must I commit to now, and what can wait?"

9. An equally important question is "What can I do to reduce the uncertainty, and how should the decision to make a more complete commitment be made?"

## III. THE STRATEGY

The case describes Fansteel, in 1963, (p. 6) as being a capital intensive producer of refractory metals, having high fixed costs, underutilization of capacity, and suffering from severe price competition in its major product line (tantalum capacitors). Mr. Hayes' strategy can be most simply interpreted as an attempt to find new markets for refractory metals, thus increasing the firm's utilization of its capacity, and to seek these new markets in areas where Fansteel can increase its value added and avoid price competition. His assumption that the fragmented fabrication industry is the barrier to more widespread use of the special metals leads him to propose forward integration into fabrication activities. Then, by bringing fabricators in contact with science and by gaining control over the whole range of fabrication processes needed to make complex structures, Fansteel will be able to work with the customers to show how to use refractory metals to solve their materials problems.

The picture Mr. Hayes and Mr. Fletcher paint of the fabrication industry is highly reminiscent of Donald Schon's description of industries that are resistant to change in his book Technology and Change:

(Industries that resist change are)

-- built on craft based rather than science based technologies;
-- fragmented, both in the sense of containing many small firms below the critical size for innovation and in the sense of dividing the work of the industry into many small steps, each under the control of separate organizations.

The result is that change tends to be limited to what each independent operator can accomplish within his own sphere and that change tends to break down when one factor in the industrial process attempts to introduce change that affects the others.

(In addition, craft oriented industries tend to)

-- focus on short-term customer service operations
-- picture itself as a job shop supplying customer demand rather than as an entrepreneur initiating new production methods and devices and marketing these to industry
-- be indifferent to research and development.

[Hayes' analysis of the fabrication industry is so similar to Schon's language that I feel he must have been influenced by this book, although he never mentioned it.]

One of the weaknesses of the Hayes-Fletcher concept of the fabrication industry is that not enough account is taken of the strong economic reasons as to why the industry is the way it is. The fabricators have generally come into existence due to a special manufacturing skill, and do not have a steadily selling product. In meeting the requirements of the aerospace industry for short production runs of complex items, it is natural that the fabricator's time horizon is short -- he must create anew his whole business every six months. The fabricators are small because of the diseconomies of scale: size decreases one's flexibility in providing customer service, increases one's vulnerability to business downturns, and vastly increases the complexity of running a job shop type business, with concomitant cost increases. The need for limited size to insure efficient short term operations, however, keeps the fabricators below the critical size necessary to support innovative development work. Until recently, new fabrication techniques were typically provided by new firms, with a type of natural selection working to keep the industry more or less up to date. Viewed in these terms, ASD's big gamble is that this process cannot keep pace with the need for solutions to pressing materials problems, and that integrated innovative firms are needed. By acquiring the most skilled fabricators early, ASD will gain an advantage over later entrants.

Hayes' background in electronics also suggests that his thinking was influenced by the early development of that field, when many small "loft" businesses developed new approaches and were quickly absorbed by firms that could market total systems to the military-aerospace buyers. It should be noted, however, that the materials industry is not like the electronics industry. The technology is not as well known, it is more capital intensive, and manufacturing, rather than design, problems have been critical.

All considered, Hayes' strategy might more properly be termed a philosophy. It is quite abstract and calls for "re-packaging" disparate fabrication skills under a systems lable and bringing together many specialties to make this new entity effective. There is no real building on corporate strengths and no firm assurance that a major opportunity exists for such a capability, even if it could be developed. This is not to say that the philosophy is poor or inappropriate, it just seems evident that there are a number of specific business strategies that could be adopted which would all fall within the overall guidelines set forth by Hayes and Fletcher.

## IV. THE ORGANIZATIONAL SUGGESTIONS

As was pointed out in section I, the key problem facing Fletcher is the design of coordinating relationships. Among the suggestions put forth by members of ASD in the case are:

Hayes: Fletcher needs a man at the division level to help him identify specific business development opportunities. Someone has to have the responsibility for tying the whole thing together (p. 16).

Holzl: What is needed to make the concept work is a group of technical people who can sell. A group is required that has the ability to invent solutions to the customer's problems on the spot and bargain over the terms of the sale; I know because I once worked in just such a group. (p. 24)

Hill: You need a phoney customer to represent the actual customer's interests. This man, or group, would have the responsibility for controlling those jobs that require the use of more than one of the plant technologies in ASD. (p. 25)

Plant Manager: Teams of engineers at the "group" level, proposed by Fletcher, would have the ability to product engineering in the areas related to the technology of their group. At lower levels the engineering talent is too limited and specialized, at a higher level the range is too broad.

Lee: What you eventually want to have is a pseudo-customer within ASD that has profit responsibility. When it gets a job requiring inputs from several plants it sub-contracts out portions of the job to those plants. It is responsible for seeing that the cost of coordination is as low as possible.

Fletcher: Two or three strong technical innovators are needed to help me. Their job is to bring ideas and opportunities to me for new projects; I will see that they are carried out.

Exhibit 1 (this note) shows the casewriter's attempt to represent the nature of the divergences among these opinions. As can be seen on the upper chart, Hill is concerned with integrating the manufacturing activities in a marketing context, while Holzl is interested in integrating the engineering activities in a marketing context. Lee seems to propose a method of manufacturing coordination that would not allow a great deal of coordination of either engineering or marketing with manufacturing. Fletcher is hard to place. He seems to want to coordinate everything but also wants to do it in a very ad hoc style. The bottom sketch shows an estimate of the relative formality of the proposals.

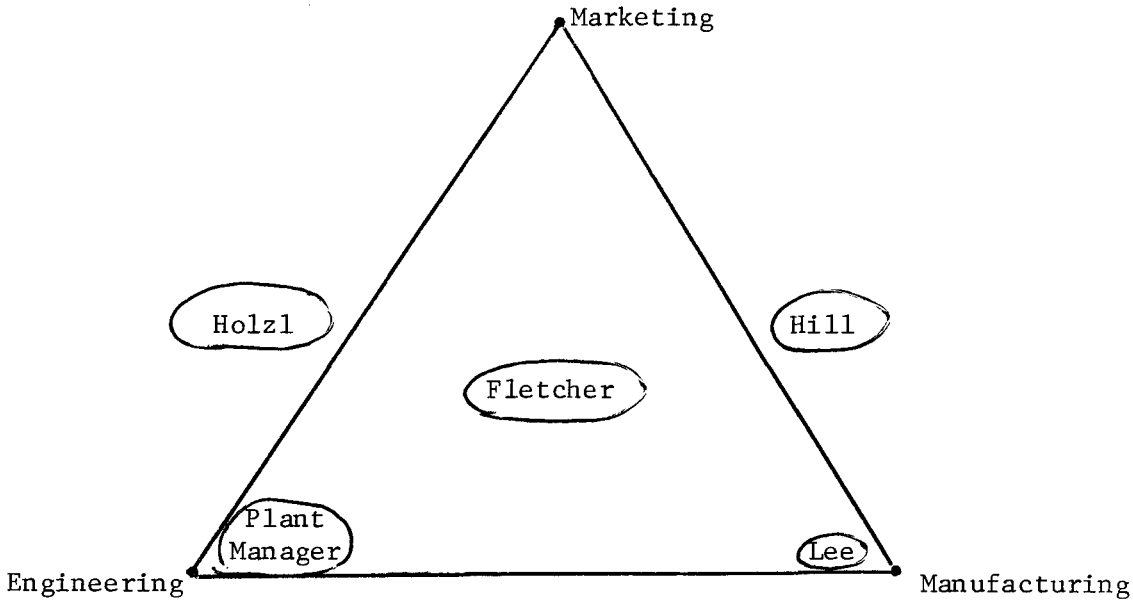
The salient point is that emphasizing marketing/engineering is not the same strategy as emphasizing marketing/manufacturing coordination. The lengthy quote by the consultant on p. 23 is relevant here. These proposals not only represent differing views on organization, they indicate differing views on the ultimate goals of the division. Combining the strategies implicit in these statements, and in those sprinkled elsewhere throughout the case, one can arrive at these two polar expressions of goals:

Strategy A: Develop a capability to engineer and design certain compound structures that require use of several fabrication capabilities. The engineering capability should extend beyond the level needed to just respond to customer requests, providing the customer with information that will help him design his product around the capabilities of ASD. Members of this (these) engineering groups should engage in some marketing activities, working closely with the customer design engineers.

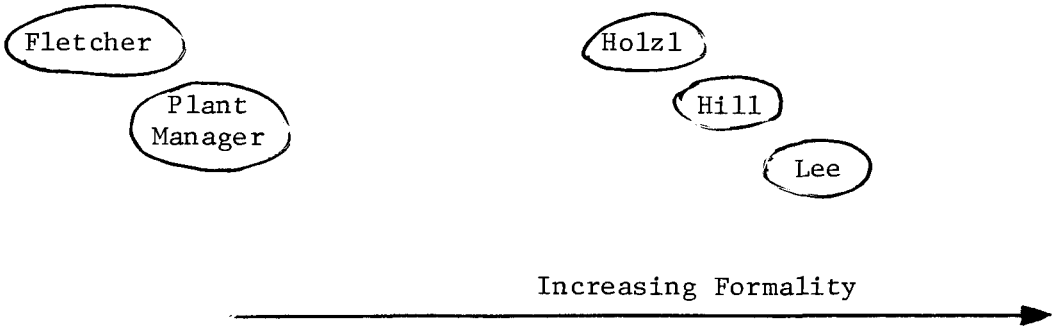
In this strategy, the premium is upon technical ability. High technology and proprietary skills means protection from price competition, so that manufacturing efficiency per se is not as vital as in strategy B.

Entreprenurial activity would focus on creating new fabrication techniques, and finding new uses for advanced materials in industry. The whole flavor of this approach is technical, with the division's ability to innovate and come up with good technical solutions to materials problems being the critical skill.

Exhibit 1




Relative emphasis of integrative activities suggested by five members of ASD




Relative formality of integrative activities suggested by five members of ASD

Strategy B: Create within ASD the ability to provide large prime contractors and equipment builders with a single-source supply of complex fabricated structures. The division would respond to a customer's need by demonstrating how ASD could do the whole manufacturing job within its own organization, while the customer might otherwise have to work with several firms.

The key tasks would be the development of the managerial skills involved in efficiently linking together the plants; doing system pricing, system marketing, bidding, contracting, systems manufacturing control, scheduling, quality control, etc., and performing this integration more efficiently than the customer could on its own. Efficiency, not technical sophistication, would be the watchword.

Thus, one of the dialogues occurring among ASD management seems to be: "If we go by Strategy A we won't need to be efficient or engage in price competition as high technology will be our competitive edge."

"Yes, but if we go by Strategy B we are avoiding the cost and risks associated with high technology and emphasize, instead, things at which we already have some ability."

## V. ORGANIZATIONAL ISSUES

In Section IV an argument was put forward that the different organizational suggestions in the case actually represent a basic conflict over the future strategy of the division. If this is true, it follows that their organizational suggestions should not be judged on the same basis, as they are proposing different means of reaching different goals. In this section, therefore, we shall investigate the organizational implications of Strategy A and Strategy B and defer to a later section the discussion of what Fletcher should do.

The Role of the Plant Managers. How do you preserve something while you change it? Fletcher (and Hayes) both maintain that the plant managers represent a large part of the value of the businesses that have been acquired. One basic conflict revealed in the case is between the advocacy of their hard-nosed attention to detail (p. 21) and the fact that these men are also "members in good standing of the fragmented fabrication industry," which is resistant to innovation (p. 22). This dilemma is one of the key issues of the case; the plant managers represent both a great opportunity and a major liability to the division.

Fletcher has recognized that running a successful fabrication job shop requires a special kind of managerial skill. There is no product line, the pressure on short term profitability is high, capital investment is high, and the daily coordination of sales activities, manufacturing, and engineering is the key task. [Would an HBS man want to do this? Would he be qualified?]

His approach to organization has therefore been designed to preserve these men as the primary actors in the division. They must, he seems to say, retain control over the sales, manufacturing, and engineering activities that they have to coordinate in order to run their businesses. Hence, the expansion of the division's capabilities must be in terms of increasing the capabilities of the sales, manufacturing, and engineering groups that they draw upon. The apparent failure of Hill's attempt to increase the centralization of the sales function (p. 20) is an indication of the plant manager's strong agreement with the principle of autonomy and Fletcher's willingness to support them. This type of conflict is at the heart of ASD's organizational problem -- how can it be better than existing industrial structure unless it actually does something differently?

While it is probably true that the men who built and ran the various shops that now make up ASD are also the most qualified to successfully operate them in the future, it is also true that there is little guarantee that many of these individuals will work for ASD as salaried managers. The manager of Airtek has already left, and Mr. Wolf and Mr. Brown may well decide to leave when their incentive agreements terminate. One could almost say that ASD can <u>not</u> afford the risk of building its organization around these men. Still, their value as managers and perhaps as entrepreneurs is considerable. One suspects that these men are, in the main, not "managers" in the sense of getting their psychic rewards from operating a business to "show" a profit; it is quite reasonable to assume that their values lean to the accumulation of wealth or to the exercise of entrepreneurial ability, or both. Since ASD plans to create a new kind of business, the entrepreneurial manager is highly valuable and, if he stays with the firm could probably be put to better use than running a single plant.

Hence we are left with these considerations:

-- The wealth oriented managers will probably leave
-- Entrepreneurial managers will be promoted or will leave
-- The operation of the plants will, therefore, pass into the hands of "management" oriented men.

It is imperative that Fletcher be prepared to have the operation of the shops pass into new hands. He should, where there is a risk of losing a plant manager, through either his leaving or promotion, insure that another man is being trained to take over the job.

<u>Groupings</u>. The grouping of the plants proposed by Fletcher would accomplish several things: it would provide a level in the organization for the more aggressive and able plant managers to move into; it would free Fletcher of the job of overseeing the operations of all the plants; it would provide overhead shelters sufficient to house new engineering teams, marketing teams, project teams, or other support activities.

Despite these advantages, the act of grouping the plants into a new profit center contradicts the policy of plant manager autonomy. A group manager without profit responsibility for operations would be put in a very awkward position -- having to defer to the plant managers on issues that they considered their concerns. [This is the position that Hill, the marketing manager is in,

one of "role conflict"] The alternative to direct profit responsibility is to give the group manager the task of managing those projects that require interplant or group-customer coordination. In this case, however, there is no real reason to call him a group manager; a title of "project manager," and a position on the division level might be more appropriate. The fact remains, nevertheless, that grouping of the plants is desirable from a purely administrative standpoint. Calling eight, or ten, or fifteen independent plants a division is surely just semantics, and not reality.

Admitting the need for group management, with profit responsibility, let us look more closely at how it might work under Strategy A and Strategy B:

> Strategy A: Here the focus is technical and the function of the groups includes the centralization of technical knowledge. An engineering team is created that can deal in all of the techniques used by the plants in the group, and develops the capability of working with both plant managers and customer engineers in trying to solve both product design and fabrication problems.
>
> Such an engineering team would be drawn in two different ways: it would support the division selling effort under Mr. Hill and would also support the plant managers within the group.
>
> The group manager would be responsible for allocating the efforts of the engineering team, would coordinate efforts on those jobs that require an integrated approach, and would be responsible for the orderly development of the group (in terms of plant projects and adding new plants).
>
> Strategy B: Here, manufacturing coordination is the key. The group manager would be responsible for coordinating the plants on those jobs that required more than one to participate. He would be the person who maintains contact with the customer, does the actual bidding, negotiates with the plant managers, allocates the profit, holds the schedule, etc.
>
> In order to do this job, the group manager would also work on improving the manufacturing management at each of the plants in his group. Grouping might lead to some economies of scale in control and permit more sophisticated job shop scheduling procedures to be adopted.
>
> The group manager would essentially function as a project manager. When jobs required group boundaries to be crossed, the division manager would assign the responsibility to one of the groups involved.

Under Strategy A, the grouping proposed by Fletcher makes the most sense, building engineering teams upon areas of common skill and knowledge. Under Strategy B, however, one would prefer the groups to be arranged so that "projects" could be undertaken without crossing group boundaries. Since projects would be market oriented, rather than technique oriented, and since the market interests of the plants all overlap, this doesn't seem possible.

The obvious resolution of these conflicts between the two strategies is to form the groups on the basis of engineering communality (as in Strategy A) and to assign the responsibility for large projects to a division level project manager. This is, however, a plan for a full blown project oriented business that is too complex for ASD to undertake immediately:

-- it would require the addition of about 10-15 highly paid personnel, an overhead burden of perhaps $500,000

-- plant manager autonomy might get lost in the shuffle

-- conflict between plant, group, and project managers would be very high in the starting phases (look what happened to the product manager idea) which would nullify any integrative gains and probably thoroughly put off the managers of prospective acquisitions.

-- The conflict between manufacturing efficiency and technical virtuosity would still exist, inflamed by the fact that the project managers, who are interested in efficiency, do not have control over the groups or plants, where group goals are in the direction of technical ability.

## VI. THE EFFECT OF STRATEGIC UNCERTAINTY

We have seen that there are two sub-strategies available to Fletcher, and that these imply different organizations and styles of management. Fletcher's real problem in choosing is that he is uncertain as to which strategy will be more effective. On p. 11 he calls for efficiency in project management and control, on p. 16 he points out the advantages of coordinating manufacturing operations rather than innovating technically, yet he also speaks of the importance of innovative engineering and suggests groups on that basis and wants to surround himself with an "elite guard" (palace guard ?) of technical people. We could assume that he is confused, but it is more useful to assume that there is real uncertainty as to which approach would be more effective in increasing profits, producing growth, and tying the division together with a unified purpose.

A commitment of resources and energy to a manufacturing - project approach when the market would respond more clearly to innovation would be a mistake, as would following a technical strategy when the barriers to real innovation might be too great. This strategic uncertainty produces a related uncertainty as to how the organization should be formed.

In addition to strategic uncertainty in terms of opportunity, Fletcher is uncertain as to how well ASD can perform as either a technically innovative or efficient manufacturing business. The skills relating to engineering and manufacturing management are different and, currently, ASD does not show evidence of excelling at either.

## VII. RECOMMENDATIONS FOR ACTION

Simply stated, Mr. Fletcher has three directions he can move in:

1. Adopt a wait and see attitude, doing only those things that are necessary for survival and exploration of the alternatives.

2. Form an organization that will be able to integrate the technical skills of the plants.

3. Form an organization that will be able to administer complex inter-plant manufacturing jobs efficiently.

Wait and See Approach. This seems to be Fletcher's current plan, as his ideas about developing ASD reflect more of an experimental spirit than those of a man tyring to build a solid organization. Given that this experimental attitude towards strategic uncertainty is reasonable, there are still several things Fletcher must do to insure the survival of the division, himself, and to permit the exploration of alternatives. They are:

1. He must keep the profitable businesses profitable and improve the performance of the losers.

2. He must begin to provide for management succession among the plant managers.

3. He must at least appear to be moving towards the materials system management concept.

4. He must increase his awareness of the nature of the market opportunities.

5. He must free himself of day-to-day concern with plant operations.

6. He must provide a capability for taking on project jobs and engaging in technical development work that is sufficient to test out these strategies.

Most of Fletcher's day-to-day problems come from those plants which are the poor performers, Airtek, Jet Engine Plant, Fansteel/Torrance, and the Powder Metallurgy Plant, and Fletcher has already stated that he plans to combine the Jet Engine Plant with Precision Sheet Metal (PSM). The only real

options he has in improving the performance of these plants is to replace the plant managers, put poor ones under the direction of better ones, and/or push the divisional marketing group to try to find more profitable work. The problems in these plants, the fact that the current central engineering group deals with just these, and the faith he has in Grien's (manager of PSM) ability in manufacturing management all suggest that putting Airtek, Fansteel/ Torrance, Jet Engine, the Powder Metallurgy Plant, and PSM together in a group headed up be Grien. If he chose to exclude Powder Metallurgy, which does have the excuse of being developmental, he would have, interestingly enough, precisely the group he labled as "metalworking." This would serve a number of purposes; it would take a great many operations problems off his shoulders, it would serve to test the group concept, and would provide a focus to see how project management could work among these plants.

Mr. Fletcher should, of course, also instruct his "entrepreneurial managers," Holzl, Brown, Preston, and Grien (p. 20) to begin training people to take over the managment of their plants.

The remaining plants would not be grouped immediately. Fletcher would work with Brown, Holzl, Preston, and Grien in seeking out opportunities for the division. As this activity was carried out, Fletcher would learn about the ability of the division to respond to various types of opportunity and plan his future organizational moves accordingly.

Technological Integration. This option stresses the leverage to be obtained by developing sophisticated technical approaches to materials problems. The key uncertainty here is whether or not this business exists in a volume sufficient to warrant a commitment of resources to the strategy.

The suggested organization is shown in Exhibit 2. Hill has been retained as the division marketing manager and will continue to carry out his present duties. In addition, he will be responsible for obtaining and making available marketing intelligence information on materials technology needs in the industries served, and will be a member of the Special Projects Committee.

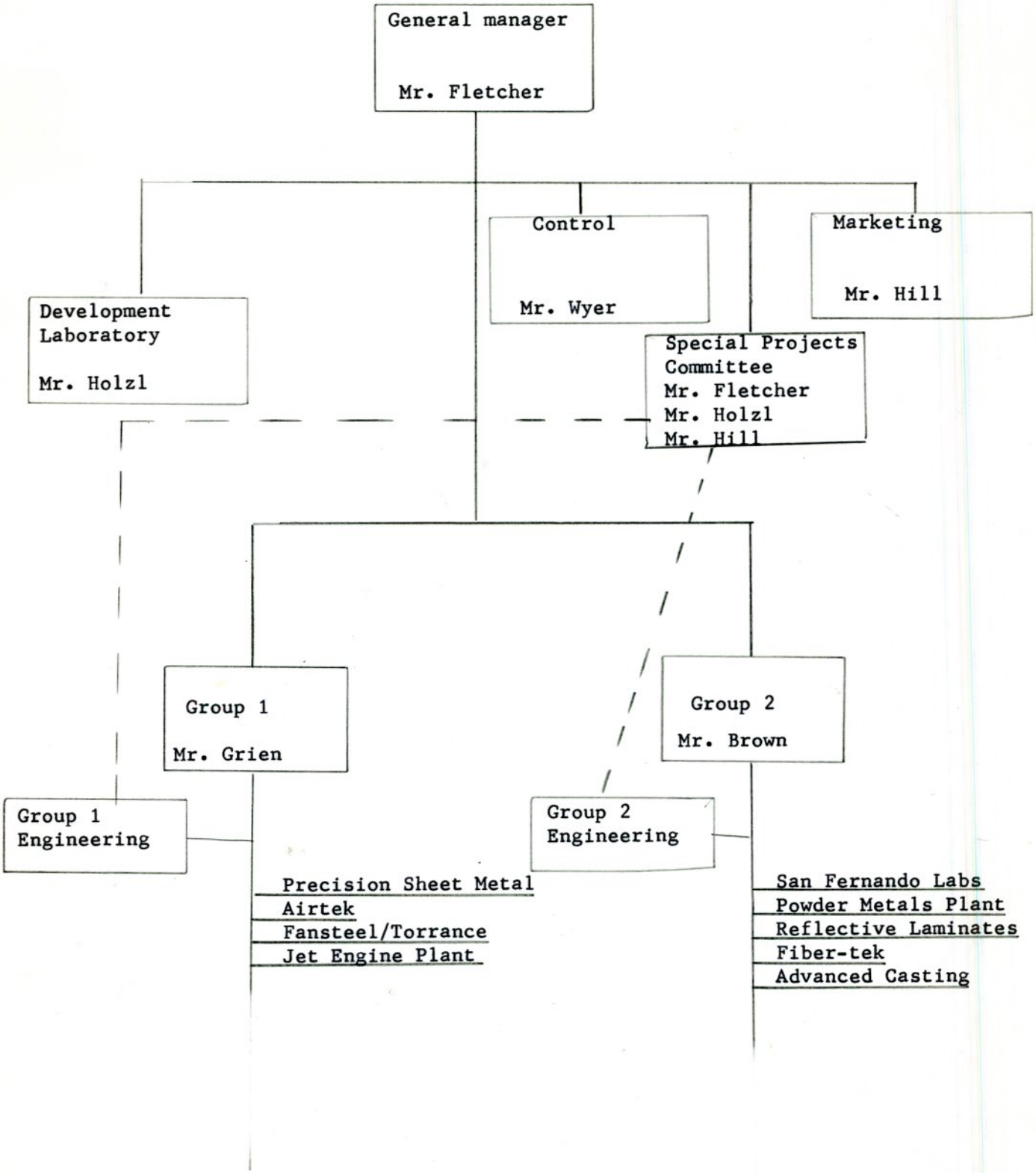
Holzl has been placed in charge of the research portion of San Fernando Labs, which has been split into R&D and manufacturing halves. His charter is to obtain as much external support for his development activities as possible and to expand into other fabrication process research and development. Holzl is also a member of the Special Projects Committee.

The third member of the Special Projects Committee is Mr. Fletcher. This committee is responsible for deciding which opportunities for creating new materials technologies will be supported and which jobs will be taken in. It also serves the marketing group by lending its members to highly technical selling, when necessary.

Mr. Grien has been put in charge of the relatively static (technically) plants and Mr. Brown in charge of the more dynamic plants. Both men have been given profit resonsibility for their respective operations

Exhibit 2

Advanced Structures Division Under Strategy A



and have been directed to work on creating a group level engineering group that will have the task of serving as technical advisor to the group manager, supervising the installation of new or improved processes, and working with the Special Projects Committee when called upon.

Mr. Wyer is in charge of the control system for the division and has been directed to develop and implement a reporting system that will provide information suitable for judging the performance of the two groups on the basis of volume, profit, costs, pricing performance, and efficiency of internal engineering teams. The two strategic pieces of information are the ability of the managers to price correctly and whether or not the engineering efforts are being justified on the basis of increased profit.

Manufacturing Integration. This option emphasizes the leverage obtained by taking over the problem of coordinating the activities of many fabricators from the aerospace prime contractors. New technologies are not stressed, efficiency is.

The key tasks are the efficient operation of the plants, the obtaining of the business of making complex structures, and the management of the project. The organizational problem is more severe than in technical integration as all unnecessary overhead must be scrupulously avoided.

The recommended organization is shown in Exhibit 3. The plants are grouped into metalworking, primary shapes, and non-metallics, with the group manager's charter emphasizing the need for effective control of manufacturing operations. The group managers will have the responsibility for multi-plant projects within their groups, with Fletcher deciding which group gets the overall responsibility in jobs that span the division.

Mr. Grien is the logical choice for the manager of metalworking, as is Mr. Brown for non-metallics. Mr. Holzl was chosen, over Mr. Preston, to head up the primary shapes group as he should be the responsible party in several projects involving the applications of San Fernando Lab's technology (his plant).

Mr. Wyer would be in charge of control and would be responsible for designing a reporting system that would allow evaluation of the group managers' performance as to profit, sales, pricing, engineering costs, and the relative cost of doing "project work." In addition, he would aid the group managers in installing their own control systems for keeping on top of project jobs.

FANSTEEL 2/24/70

TEACHING OUTLINE

① What is this concept of mat. sys. mgt. (MSM)?
How is MSM related to nature of mat. fab. industry?

② What are the critical tasks associated with MSM?
How do these differ from the current success requirements of most mat. fab. shops?

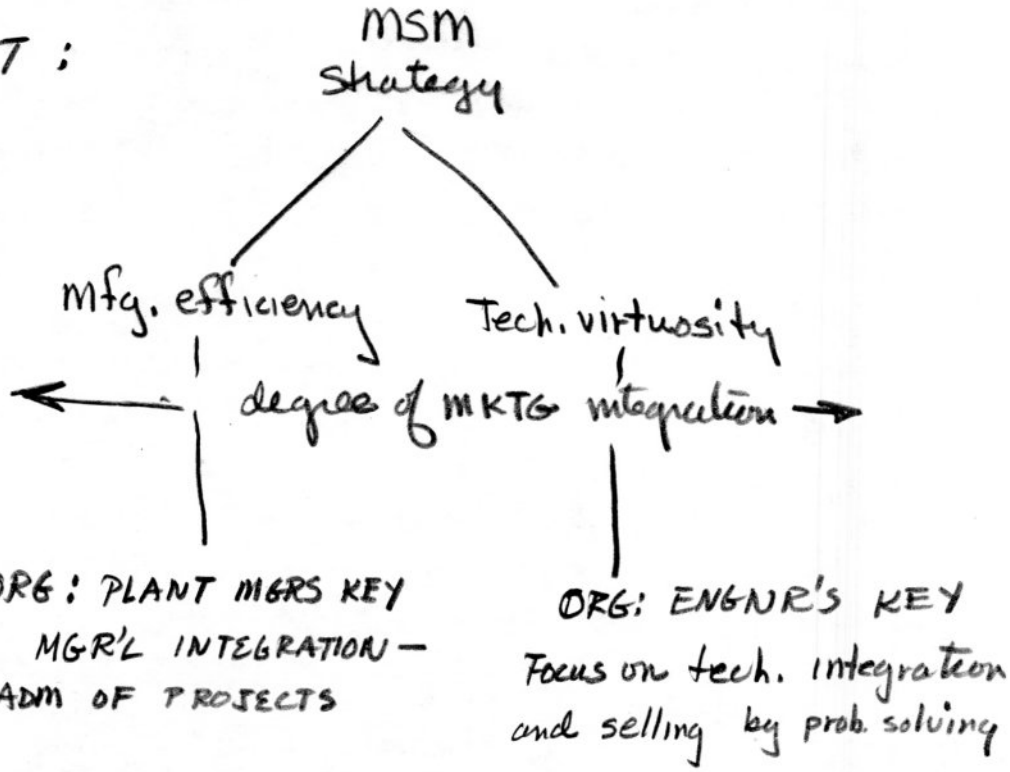
③ What is the degree of Risk in this strategy?

④ What org. alternatives are evident in case?
To what extent do they fulfill the needs of fab. shops and MSM?

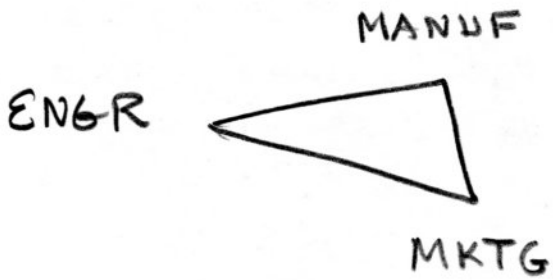
Do these org. alts. represent different ways of achieving the same goal, or ways of achieving different goals?
Does the consultants summary on pp 23 describe the same org. alts?

(6) Is Fletcher facing an organizational or a strategic problem?

DRAW THIS FIRST:




STRATEGY ⟺ STRUCTURE




INTEGRATION VS. PLANT MGR AUTONOMY

BOARD 1

## Mat Sys Mgt

tlet for Fur-steel metals
n science base of fabr's
iede single source supplier
new product concepts
y comb. fab techs.

R UTILITY + MARKETABILITY
FAB TECH.

ISK = GAMBLE

## MSM TASKS

Relieve Fletcher of day-to day tasks
Plant mgr autonomy
Profit ind. plants
low cost coord
Int. engrs.
Bring tech to cust - solve problems
Control whole mfg-engrs sequence

## FAB SHOP TASKS

Efficient prod
Cust. serv.
flexibility
Weather cycles — Style 1
Skilled workers
Low overhead
Strong manager

engineer to order
special tech.
dev new fab techniques — Style 2
dev new pdcts

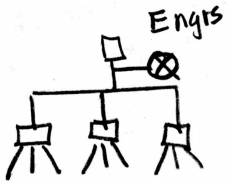
## Fab. Industry

Fragmented by fab tech
Many small job shops
Cyclic business
More craft than science
Few large customers
Economic Darwinism — method of change

Incr need for adv. materia
and mat. sys.

BARD 2

| ORG ALT | PRO | CON |
|---|---|---|
| Fletcher's plan<br>group by fab tech<br>elite guard | provides overhead umbrella's<br>releases fletcher<br>centralizes tech knowledge<br>provides single pt of contact | decr plant MGR cont.<br>+ stakes<br>no prov for project adm.<br>where find men who know enough |
| group by tech static<br>tech changing | more eff overhead base<br>cross tech boundaries | mix dis-similar manuf. |
| Product MGR | integrate MKTG ACT<br>central prod knowledge | Plant mgrs don't want<br>day to day interference<br>MKTG doesn't understand tech |
| CUST oriented<br>WHIZ KIDS | new app + cust | mfg coord weak<br>no adm. provision<br>where get men? |
| Profit resp Project Office | mfg int.<br>rep total cup to cust.<br>adm project | may threaten plant<br>mgr autonomy<br>poorer job at tech int<br>+ cust prob solving. |



(7) Summary of Fletcher's situation:

He has a philosophy (msm)

He faces a range of possible org. actions

PORTFOLIO
MFG INT
TECH INT
MKTG INT

} as risk decreases so does ability of structure to sieze opportunity

RISK MGT problem due to <u>STRATEGIC</u> <u>uncertainty</u>

STRAT. UNCERTAINTY ⟹ STRUCT. UNCERTAINTY

<u>Yet</u>, he must provide for economic operations and at the very least provide the org. with the means to experiment & explore.

(8) What would you advise Fletcher to do?

a) Provide for mgt. succession

b) focus on most promissing pdct-mkt area
spec. proch. group at div level (2 men?) to coord MKTG + engrg. No profit resp. until org learns to work in this manner.

+ ?